DGUV Test Prüf- und Zertifizierungsstelle Elektrotechnik Fachbereich Energie Textil Elektro Medienerzeugnisse	**Test Protocol** **Laser protection wall** **ASSA ABLOY Entrance Systems**	Az.: 13-166 Attachment: 4 Trial: 4 Name: Gomolka Page: 1 of 2 Date: 15 Oct. 2013

Client: ASSA ABLOY Entrance Systems
Am Mondschein 25
59557 Lippstadt, Germany

Participants: Mr. Frede (ASSA ABLOY), Mr. Bocksrocker (Trumpf) and Mr. Gomolka (Electrical engineering testing facility)

Test location: TRUMPF Laser- und Systemtechnik GmbH, 71254 Ditzingen, Germany

Test date: 15 Oct. 2013

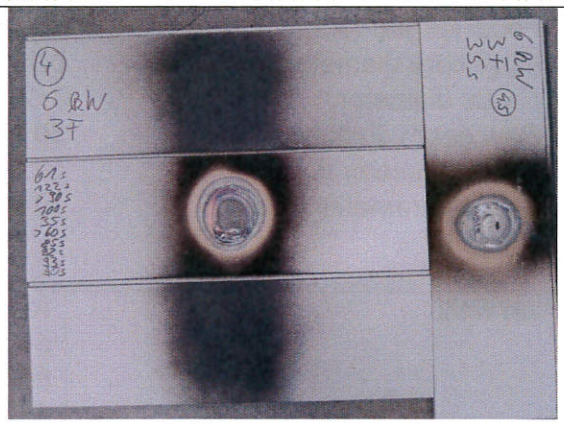
Test specifications: Test specification PL-Info 007 from 17 Jun. 2013

Laser specifications:

- Manufacturer: Trumpf TruDisk 8002
- Wavelength: 1030 nm
- Optics: Trumpf PFO 3D
- Focal length: 450 mm
- Fibre core diameter: 0.2 mm
- Focus diameter: 0.65 mm
- Raw beam diameter: 30 mm
- Beam diameter d_{63}: 19.5 mm
- Beam-parameter-product: 10 mm x mrad

Protection wall:	RR 3000, standard lamella, No. 4707R0044, aluminium anodized, 1.6/1.6_20_125 mm, AlMgSi 0.5 F25 with lamellar seal EPDM No. 4401R0014
Specimen size:	300 x 300 mm (3 lamellae)
Angle of incidence:	90° ± 3° (lamella centre)
Distance:	3F
Laser output:	6.0 kW
Focal spot diameter: (measurement/calculation)	59 mm
Radiation intensity at the test surface:	2.23 x 10^6 W/m^2

DGUV Test Prüf- und Zertifizierungsstelle Elektrotechnik Fachbereich Energie Textil Elektro Medienerzeugnisse	**Test Protocol** **Laser protection wall** **ASSA ABLOY Entrance Systems**	Az.: 13-166 Attachment: 4 Trial: 4 Name: Gomolka Page: 2 of 2 Date: 15 Oct. 2013

Pre-treatment of the test surface: (synthetic resin-based spray paint, jet black matt, RAL 9005)	Yes
Protection wall ventilation:	Central extraction system in the test cell and a portable weld fume extraction unit at a lateral distance of approx. 30 cm from the specimen (sensing element cross-section 200 x 500 mm)
Indicator:	Thermal paper at a distance of 200 mm
Measured values in s: (up to penetration of the protection wall)	61, 122, >90, 100, 35, >60, 85, 92, 93, 90
Duration of protection in s: (lowest measured value x 0.7)	**25**
Remarks	

Malte Gomolka
Test engineer

DGUV Test Prüf- und Zertifizierungsstelle Elektrotechnik Fachbereich Energie Textil Elektro Medienerzeugnisse	**Test Protocol** **Laser protection wall** **ASSA ABLOY Entrance Systems**	Az.: 13-166 Attachment: 5 Trial: 5 Name: Gomolka Page: 1 of 2 Date: 15 Oct. 2013

Client: ASSA ABLOY Entrance Systems
Am Mondschein 25
59557 Lippstadt, Germany

Participants: Mr. Frede (ASSA ABLOY), Mr. Bocksrocker (Trumpf) and Mr. Gomolka (Electrical engineering testing facility)

Test location: TRUMPF Laser- und Systemtechnik GmbH, 71254 Ditzingen, Germany

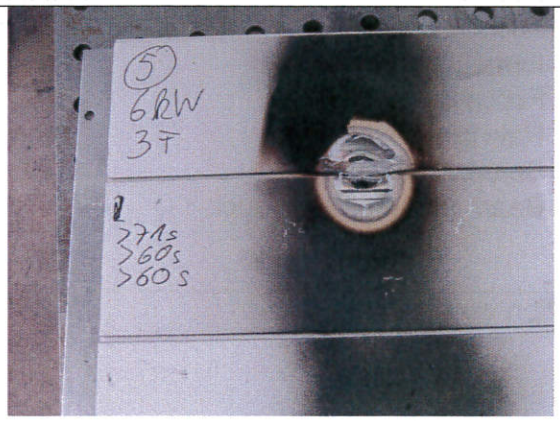
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Test specifications: Test specification PL-Info 007 from 17 Jun. 2013

Laser specifications:

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- Focal length: 450 mm
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- Raw beam diameter: 30 mm
- Beam diameter d_{63}: 19.5 mm
- Beam-parameter-product: 10 mm x mrad

Protection wall:	RR 3000, standard lamella, No. 4707R0044, aluminium anodized, 1.6/1.6_20_125 mm, AlMgSi 0.5 F25 with lamellar seal EPDM No. 4401R0014
Specimen size:	300 x 300 mm (3 lamellae)
Angle of incidence:	90° ± 3° (in the area of the seal)
Distance:	3F
Laser output:	6.0 kW
Focal spot diameter: (measurement/calculation)	59 mm
Radiation intensity at the test surface:	2.23×10^6 W/m^2

Pre-treatment of the test surface: (synthetic resin-based spray paint, jet black matt, RAL 9005)	Yes
Protection wall ventilation:	Central extraction system in the test cell and a portable weld fume extraction unit at a lateral distance of approx. 30 cm from the specimen (sensing element cross-section 200 x 500 mm)
Indicator:	Thermal paper at a distance of 200 mm
Measured values in s: (up to penetration of the protection wall)	>71 >60 >60
Duration of protection in s: (lowest measured value x 0.7)	25 (the more restrictive value from Trial 4 applies)
Remarks:	

Prüf- und Zertifizierungsstelle DGUV Test Fachbereich ETEM-ET

Malte Gomolka
Test engineer